Hannah Christina Arjonillo

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Currently a Physics PhD student at UP Diliman with research interests in sociophysics and complex systems, and independently pursuing AI safety research. 6+ years of AI/ML industry experience in medtech and healthcare, spanning physiological time-series ML, behavioral text topic modeling, and health outcomes modeling.

Experience

Research Fellow, Supervised Program for Alignment Research (SPAR), Fall 2026 (Ongoing)
Remote | Sep 2026 – Present

- Project: "Loss of Human Agency in the Age of Advanced AI – An Agent-Based Model," mentored by Zhamilia Klycheva, Ph.D.

Data Scientist, BioIntellisense Inc.
Remote / Golden, CO | Dec 2024 – Apr 2026

- Led the development of a core body temperature estimation algorithm from skin and ambient temperature signals; supported clinical trial study design and data collection through data quality checks, literature reviews, and model iterations
- Benchmarked a vendor temperature model against the internal model using Bland-Altman and correlation analyses on clinical trial data; conducted error analyses on study limitations, informing study design and vendor discussions
- Contributed to clinical outcomes analyses on large-scale EMR data — built propensity-score-matching pipelines with calipered nearest-neighbor matching, Poisson log-link regressions for mortality and length-of-stay, subgroup interaction models, and balance diagnostics across hospital and hospital-to-home cohorts
- Developed candidate algorithms for wearable on/off-body detection and binary sleep/wake classification using multimodal signals (HR, HRV, skin temperature, accelerometry, ECG contact quality)
- Contributed to the team's clinical alerting system, including analysis of vitals alarm thresholds (heart rate, respiratory rate) to balance detection sensitivity against alarm fatigue in continuous patient monitoring
- Analyzed ECG-PPG pulse arrival time for cuffless blood pressure estimation, building subject-wise models and limits-of-agreement evaluations
- Constructed reusable ICD/DRG/MDC and Elixhauser comorbidity mappings from CMS reference tables for confounding control across analyses

Tools: Python, Databricks, MLflow, Claude Code

Machine Learning Scientist, Sequelae PH Inc. / Lore Health LLC
Remote / San Diego, CA | Dec 2022 – Nov 2024

- Developed and productionized embeddings-based topic models that aggregate human-generated content from the platform (Lore Health app / community.lorehealthcare.com) into featured central ideas called Lore
- Built text features distinguishing reflection from plain observation. Grounded the approach in construal-level theory from the psychology literature and operationalized it with published word-level concreteness ratings (Brysbaert, 2013), a more cost-effective and deterministic alternative to prompting an LLM
- Built and iterated on related models describing narrative types to inform how conversations evolve, e.g. whether a passage shares a personal experience or solicits others' experiences

- Aided in the development of human-in-the-loop processes, including automating manual-file-to-SQL-Server transfers, collaborating with software developers and writers, and working with Snorkel (annotation and modeling)
- Contributed to internal text processing and cloud utility packages

Tools: Python, Microsoft SQL Server, Google Cloud Platform, Airflow, Docker, REST API, Snorkel, Backstage, OpenAI API

Insighting Expert, Advanced Analytics Division, Enterprise Data Office, Globe Telecom, Inc.
Taguig City, Philippines | Jun 2022 – Dec 2022

- Developed an algorithm to spatially aggregate cell site activity and features based on their longitude and latitude
- Produced Tableau dashboards and presented spatial analysis findings to senior leadership

Tools: Python, Tableau

Lecturer (Part-time), National Institute of Physics, University of the Philippines - Diliman
Diliman, Quezon City, Philippines | Feb 2022 – May 2022

- Taught introductory physics and laboratory courses on electromagnetism to undergraduate students online

Data Scientist, OptumLabs, Inc. / UnitedHealth Group Research and Development (UHG R&D)
Cebu City, Philippines | Jan 2020 – Mar 2022

- Used NLP techniques to explore user reviews on a particular app acquired by UnitedHealth Group
- Developed a classification model that uses economic factors and health factors to predict whether a patient will soon develop Type 2 Diabetes, and compared its results to existing literature
- Contributed to the development of an epidemiological model (SIRD) to predict the cost of COVID-19 testing and the corresponding strategy to optimize the cost based on location infection rates, and developed the backend for a public-facing COVID-19 testing strategy simulator
- Investigated non-standard requests and problems using equation-based modeling, machine learning methods, statistics, and network science, implemented in Python and/or R
- Created reports, processed tables, and/or interactive dashboards to provide resources and tools to internal and external stakeholders

Tools: Google BigQuery, Python, R, Dash

Data Scientist Apprentice, UnitedHealth Group Research and Development (UHG R&D)
Makati, Philippines | Jul 2019 – Dec 2019

- BPI-UHG Data Science (BUDS) Training Program
- Acquired foundations in statistics, machine learning, and programming for data science research

Tools: Google BigQuery, Python, R, Tableau

Skills

- **Programming:** Python, R, Arduino C/C++
- **Data engineering:** Databricks, Google BigQuery, Microsoft SQL Server, Airflow
- **Model deployment:** FastAPI, Docker, Google Cloud Run
- **Dashboard deployment:** Streamlit, Dash (Python), CSS
- **AI-assisted workflows:** Claude Code, OpenAI API
- **Development:** GitHub / git, Jupyter Notebook, MLflow, Snorkel, PyCharm, RStudio, Backstage
- **Office:** LaTeX, Microsoft Office, Google Docs Suite

Publications

- Euste, J. L., Arjonillo, H. C. and Saloma, C. (2024) Time-resolved culture maps derived from the Integrated Values Survey data (1981-2022). **Physica A: Statistical Mechanics and its Applications 659:130317**, https://doi.org/10.1016/j.physa.2024.130317
- Arjonillo, H. C. and Saloma, C. (2023) Monte Carlo analysis of focused Gaussian beam in scattering media: Curvature correction and Mie scattering. **Optics Communications 535:129314**, https://doi.org/10.1016/j.optcom.2023.129314. International Publication Award (IPA) 2023 recipient, University of the Philippines - Diliman

Education

- **University of the Philippines - Diliman**, PhD Physics (August 2025 - Present). Research focus: Hyperspectral unmixing applied to senatorial voting patterns in the Philippines. Adviser: May T. Lim, PhD
- **University of the Philippines - Diliman**, MS Physics (January 2021 - June 2024). Thesis title: Measuring intra-country cultural diversity. Adviser: Caesar A. Saloma, PhD
- **University of the Philippines - Diliman**, BS Applied Physics (Instrumentation Physics), Cum Laude (August 2014 - June 2019). Thesis title: Towards a more accurate Monte Carlo analysis of propagating focused Gaussian beam in scattering media. Adviser: Caesar A. Saloma, PhD

Conferences, Workshops, & Courses

- **BlueDot Impact: Biosecurity** (October 2026 - Present) — In progress. Participant in a part-time cohort-based course.
- **BlueDot Impact: Technical AI Safety** (October 2026) — Completed this course. Certificate: https://bluedot.org/certification?id=rec8tl7Xhritezg4X
- **BlueDot Impact: AGI Strategy** (September 2026) — Participant in a cohort-based course. Certificate: https://bluedot.org/certification?id=recCymmBqHu60f4RZ
- **BlueDot Impact: Future of AI** (August 2026) — Completed this self-paced course. Certificate: https://bluedot.org/certification?id=recsJMS8iet30UCLD
- **42nd Samahang Pisika ng Pilipinas Physics Conference** (June 2024, Online) — Contributor and presenter. H. C. Arjonillo, J. L. Euste, and C. Saloma. Measuring cultural diversity within countries with ellipses, contour maps, and distances, Proceedings of the Samahang Pisika ng Pilipinas 42, SPP-2024-1D-01 (2024). https://proceedings.spp-online.org/article/view/SPP-2024-1D-01
- **Workshop on Quantitative Human Ecology, Abdus Salam International Centre for Theoretical Physics** (July 2022, Online) — Participant. Workshop bringing together scientists from economics, ecology, demography, physics, complex systems science, climatology, epidemiology, social sciences, and intelligent systems to promote wider integration of methods across disciplines tackling sustainability challenges.
- **39th Samahang Pisika ng Pilipinas Physics Conference** (October 2021, Online) — Contributor and presenter. H. C. Arjonillo, J. L. Euste, and C. Saloma. Correlating cultural values with COVID-19 outcomes, Proceedings of the Samahang Pisika ng Pilipinas 39, SPP-2021-PB-20 (2021). https://paperview.spp-online.org/proceedings/article/view/SPP-2021-PB-20
- **37th Samahang Pisika ng Pilipinas Physics Conference** (June 2019, Bohol, Philippines) — Contributor and presenter. H. C. Arjonillo and C. Saloma. Curvature correction and Mie scattering in Monte Carlo analysis of propagating focused Gaussian beam in scattering media, Proceedings of the Samahang Pisika ng Pilipinas 37, SPP-2019-PB-20 (2019). https://paperview.spp-online.org/proceedings/article/view/SPP-2019-PB-20
- **36th Samahang Pisika ng Pilipinas Physics Conference** (June 2018, Palawan, Philippines) — Contributor and presenter. H. C. Arjonillo and C. Saloma. Preserving photon detection and average

power in random scattering media and curvature correction in vectorized particle Monte Carlo simulation of focused Gaussian beam, Proceedings of the Samahang Pisika ng Pilipinas 36, SPP-2018-2C-06 (2018). https://paperview.spp-online.org/proceedings/article/view/SPP-2018-2C-06

Music Education

- **Associated Board of the Royal Schools of Music (ABRSM)**, Diploma in Music Performance (Violin) (2016)
- **Associated Board of the Royal Schools of Music (ABRSM)**, Grade 8: Violin (Distinction), Piano (Merit) (2014)
- **Associated Board of the Royal Schools of Music (ABRSM)**, Grade 5: Theory of Music (Merit) (2012)